

Steps	Reason
1) $p \rightarrow q$	Premise
2) $p \rightarrow q$	Premise
3) $\neg q$	Premise
4) $\neg p$	Modus tollens on 3, 1
5) $\neg p$	Modus tollens on 3, 2

Valid conclusions: 1) I did not eat spicy food 2) There is no thunder while I sleep.

c) "I am either clever or lucky." "I am not lucky." "If I am lucky, then I will win the lottery."

Soln: ~~p~~ p: I am clever q: I am lucky r: I will win the lottery.

Premises: 1) $p \vee q$ 2) $\neg q$ 3) $q \rightarrow r$

Steps	Reason
1) $p \vee q$	Premise
2) $\neg q$	Premise
3) $q \rightarrow r$	Premise
4) p	Disjunctive syllogism on 1, 2

Valid conclusion: I am clever

d) "Every computer science major has a personal computer." "Ralph does not have a personal computer." "Ann has a personal computer."

Soln: ~~CS(x)~~ $CS(x)$: x is a computer science major $PC(x)$: x has a personal computer.

Premises: 1) $\forall x (CS(x) \rightarrow PC(x))$ 2) $\neg PC(\text{Ralph})$ 3) $PC(\text{Ann})$

Steps	Reason
1. $\forall x (CS(x) \rightarrow PC(x))$	Premise
2. $\neg PC(\text{Ralph})$	Premise